Lecture 14 — 27.11.25

Galerkin methods

So far, we have only reformulated the problem, but where is the (numerical) approximation?

$\rightsquigarrow$ general Galerkin strategy (similar in numerical linear algebra, e.g., CG)

(Known) **Step 1** Find the variational/weak form:

$$\text{Find } u \in V \text{ s.t. } a(u,v) = F(v) \quad \forall v \in V. \qquad (*)$$

Trial and test space do not need to coincide ("Petrov-Galerkin") but often do.

(New) **Step 2** Perform a Galerkin dimension reduction: For each $n \in \mathbb{N}$ let $V_n \subset V$ be a subspace of V with $\dim(V_n) = n$. Then solve (as approximation to $(*)$):

$$\text{Find } u_n \in V_n \text{ s.t. } a(u_n, v_n) = F(v_n) \quad \forall v_n \in V_n. \qquad \text{(Galerkin equation)}$$

(New) **Step 3** Derive a linear system of equations: note, V_n is finite-dimensional and we can find a basis $\{\varphi_1, \dots, \varphi_n\}$ of V_n. Moreover, each element in V_n, in particular the solution u_n of the Galerkin equation can be written as

$$u_n = \sum_{i=1}^{n} \alpha_i \varphi_i \quad \rightsquigarrow \text{ but how do we get the coefficients } \alpha_i?$$

$\rightsquigarrow$ since a is bilinear, from the Galerkin equation we obtain

$$(1) \quad a(u_n, v_n) = a\Big(\sum_{i=1}^{n} \alpha_i \varphi_i, v_n\Big) = \sum_{i=1}^{n} \alpha_i\, a(\varphi_i, v_n) \overset{\text{if } v_n = \varphi_j}{=} \sum_{i=1}^{n} \alpha_i\, a(\varphi_i, \varphi_j)$$

Instead of testing with arbitrary v_n in the Galerkin equation, focus on $v_n = \varphi_j$:

$$(2) \quad \text{Find } u_n \in V_n \text{ s.t. } a(u_n, \varphi_j) = F(\varphi_j) \quad \text{for all } j \in \{1, \dots, n\}.$$

Combining (1) and (2), we obtain n equations for n unknowns $\alpha_1, \dots, \alpha_n$:

$$A_n \alpha := \underbrace{\begin{bmatrix} a(\varphi_1,\varphi_1) & \cdots & a(\varphi_n,\varphi_1) \\ \vdots & \ddots & \\ a(\varphi_1,\varphi_n) & \cdots & a(\varphi_n,\varphi_n) \end{bmatrix}}_{\text{Galerkin matrix } A_n} \underbrace{\begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}}_{=:\alpha} = \underbrace{\begin{bmatrix} F(\varphi_1) \\ \vdots \\ F(\varphi_n) \end{bmatrix}}_{=:f_n} \in \mathbb{R}^n$$

Note testing "only" with basis functions φ_j instead of arbitrary $v_n \in V_n$ is <u>no</u> restriction

Indeed, consider an arbitrary $v_n \in V_n \Rightarrow$ has a basis representation $v_n = \sum_{j=1}^{n} \beta_j \varphi_j$, with some coefficients $\beta_j \in \mathbb{R}$

$$\Rightarrow \quad a(u_n, v_n) = \sum_{i=1}^{m} \alpha_i \, a(\varphi_i, v_n) = \sum_{i=1}^{n} \alpha_i \left(\sum_{j=1}^{n} \beta_j \, a(\varphi_i, \varphi_j) \right)$$

$$= \sum_{j=1}^{n} \beta_j \left(\sum_{i=1}^{m} \alpha_i \, a(\varphi_i, \varphi_j) \right) = \sum_{j=1}^{n} \beta_j \, a(u_n, \varphi_j) = \sum_{j=1}^{n} \beta_j \, F(\varphi_j)$$

$$\overset{F \text{ is linear}}{=} F\left(\sum_{j=1}^{n} \beta_j \varphi_j \right) = F(v_n)$$

Galerkin orthogonality and best approximation

For the solution u of the variational problem we know that

$$a(u, v) = F(v) \quad \text{for all } v \in V$$

Since by construction $V_n \subset V$, this in particular implies

$$a(u, v_n) = F(v_n) = a(u_n, v_n) \qquad \text{for all } v_n \in V_n \subset V$$

Bilinearity of a yields then

$$a(u - u_n, v_n) = 0 \qquad \text{for all } v_n \in V_n \qquad \text{Galerkin orthogonality}$$

The error $u - u_n$ is "a-orthogonal" to all vectors from V_n

Note/recall orthogonality is related to best approximation (see later on)

Definition Let V be an $\mathbb{R}$-vector space and $a: V \times V \to$ be a bilinear form. Then we call a

- <u>symmetric</u>, if $a(u, v) = a(v, u) \quad \forall u, v \in V$
- <u>positive definite</u>, if $a(u, u) \geq 0$ for all $u \in V$ and $a(u, u) = 0$ if and only if $u = 0$
- <u>negative definite</u>, if $-a$ is positive definite

$\rightsquigarrow$ allows us to obtain unique solutions for the variational problem if a is pos. def.

Indeed, consider two solutions u_1, u_2 of $a(u, v) = F(v)$ for all $v \in V$.

Then $a(u_1, v) = F(v) = a(u_2, v)$ for all $v \in V$.

Hence $a(u_1 - u_2, v) = 0$ for all $v \in V$, in particular for $v = u_1 - u_2 \in V$.

$\Rightarrow \quad a(u_1 - u_2, u_1 - u_2) = 0 \quad \Rightarrow u_1 - u_2 = 0$ if a is positive definite

$\Rightarrow$ solution is unique!

<u>Example</u> (choice of vector space influences positive definiteness!)

Consider

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v \, dx$$

as a bilinear form $a: V \times V \to \mathbb{R}$ with two different choices for V:

1) $V = \{ v \in H^1(\Omega) \mid \frac{\partial v}{\partial \nu} = 0 \text{ on } \partial\Omega \}$

$$\Rightarrow \quad a(u,u) = \int_\Omega \nabla u \cdot \nabla u \, dx = \int_\Omega \|\nabla u\|^2 \, dx$$

If $a(u,u) = 0$, then $\|\nabla u\| = 0$ but this does not imply $u = 0$!

Indeed, $u(x) = 1$ for all $x \in \Omega$ satisfies $u \in H^1(\Omega)$, $\frac{\partial u}{\partial \nu} = 0$ on $\partial\Omega$

so that $u \in V$ but $u \neq 0$ $\Rightarrow$ a is not positive definite over V

2) $V_0 = \{ v \in H^1(\Omega) \mid v = 0 \text{ on } \partial\Omega \}$

Again, for $u \in V_0$ we first conclude $\nabla u = 0$ if $a(u,u) = 0$.

However, if a function u satisfies $\nabla u = 0$, it is constant over Ω and

if it is in V_0, it also satisfies $u = 0$ on $\partial\Omega$ $\Rightarrow$ $u(x) = 0$ for all $x \in \Omega$!

$\Rightarrow$ a is positive definite over V_0!